

HARSH RAJ

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EDUCATION

Northeastern University, Boston, MA	July 2026 (expected)
Master of Science in Computer Science	GPA: 3.6/4.0
Delhi Technological University, Delhi, India	August 2023
Bachelor of Technology in Engineering Physics	GPA: 3.4/4.0

PROFESSIONAL EXPERIENCE

Research Assistant, Forschungszentrum Jülich	July 2025 – Present
- Contribute to EU Open Foundational Model projects (MINERVA, OpenEuroLLM, ELLIOT) on JUPITER, the EU's largest supercomputer - Lead ablation studies on dataset mixtures and scaling laws on LLM training data at the petabyte scale - Optimized multi-node data processing pipelines using Ray, achieving 5x speedup in deduplication, filtering and map reduction operations	
Co-Founder, Ontocord AI (Open Source)	Jan 2024 – Present
- Co-founded open-source initiative focused on creating high-quality permissive datasets for LLMs - Developed data processing pipelines with strategies for mixing, formatting, upsampling, interleaving, and quality validation - Published MixtureVitae-300BT, the state-of-the-art permissive pretraining dataset to date, achieving 10% performance improvement over existing permissive datasets while matching non-permissive alternatives. Work under review at ICLR 2026.	
Applied Scientist, Intern, Vijil AI	Feb 2024 – Nov 2024
- Contributed to the vijil-evaluation platform for red-teaming of enterprise AI agent workflows. Integrated various benchmark datasets for vulnerability assessment - Developed the vijil-fuzzer framework for adversarial text mutation for red-teaming. It achieved a 80% success rate against GPT-4o at launch	
Machine Learning Engineer, Yield Inc	Sept 2022 – Jan 2024
- Built synthetic data generation and finetuning infrastructure for LLM agents - synchaev and mandrill - Co-developed Cacti, a Web3 transaction chatbot that simplifies blockchain interactions through natural language	

PUBLICATIONS

Improving Consistency in Large Language Models through Chain of Guidance 1st Author	TMLR 2025
Defences against Reverse Preference Attacks 3rd Author	AAAI 2025
Transfer of Adversarial Robustness from Pretraining to Downstream 2nd Author	NeurIPS 2023
Measuring Reliability of Large Language Models through Semantic Consistency 1st Author	NeurIPS 2022
<i>Best Paper Award</i>	

PROJECTS

terminal-bench (600+ GitHub ★)	Language agent sandbox evaluation framework (Co-author & Maintainer)
inspect-ai (1300+ GitHub ★)	Industry-standard LLM evaluation suite (Contributor)

AWARDS AND ACHIEVEMENTS

SWISS AI Grant Recipient – Awarded 200k GPU hours and CHF 61,000 in funding	2025
Vision and Language Navigation : Secured 3rd position on the R2R benchmark	2024
Rank 7 – Codeforces Raif ML Round 1 Competition	2021